

DATA STORYTELLING



TOTAL SALES

139.40M



TOTAL ORDERS

992



TOTAL CUSTOMERS

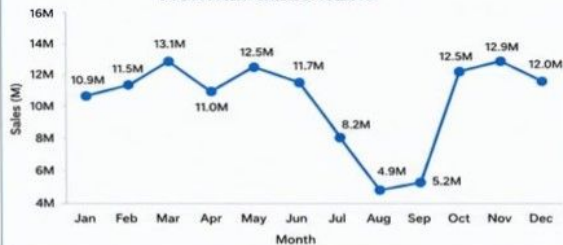
947



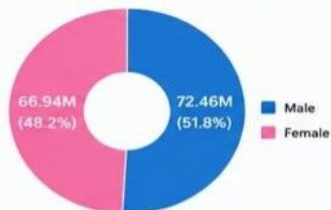
TOP CATEGORY

Electronics

MONTHLY SALES TREND



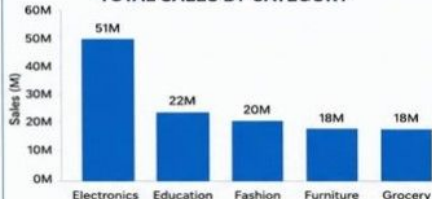
TOTAL SALES BY GENDER



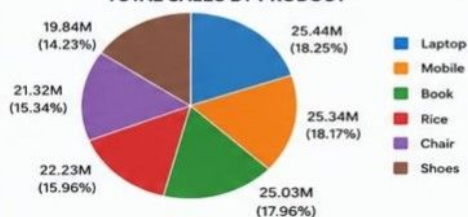
KEY INSIGHTS

- ▶ Sales showed fluctuations throughout the year, with a noticeable dip in August–September and a strong recovery in the final months.
- ▶ **Electronics** is the top-performing category and contributed the highest share of total sales.
- ▶ Sales contribution from male and female customers was nearly balanced, indicating a diversified customer base.
- ▶ Product-wise sales were fairly evenly distributed, showing that revenue was not dependent on only one product.
- ▶ Strong sales performance in March, June, and October–December acted as major growth periods.

TOTAL SALES BY CATEGORY



TOTAL SALES BY PRODUCT



KEY REVENUE DRIVERS

- ✓ Electronics category was the top revenue contributor.
- ✓ Laptop and Mobile were among the strongest-selling products.
- ✓ March, June, and October–December were the best-performing sales months.
- ✓ Male customer segment contributed slightly higher sales overall.



CONCLUSION

Overall sales performance remained stable despite monthly fluctuations. Electronics emerged as the strongest revenue-driving category, while category, product, and gender-wise sales remained fairly balanced, indicating a diversified revenue structure. The strong recovery in the final months reflects positive business momentum and improving sales performance.



GOAL

To analyze sales performance across months, categories, products, and customer gender in order to identify key revenue drivers, sales trends, and overall business performance patterns.

DATA STORYTELLING

CUSTOMER DISTRIBUTION ANALYSIS



TOTAL CUSTOMERS

947



TOTAL ORDERS

992



TOTAL SALES

139.40M



AVG ORDER VALUE

140.52K



MALE CUSTOMERS

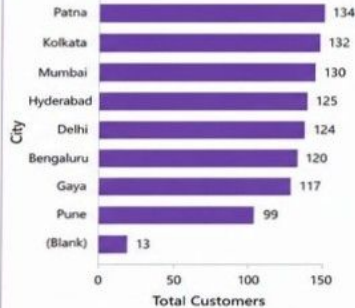
479 (48.83%)



FEMALE CUSTOMERS

502 (51.17%)

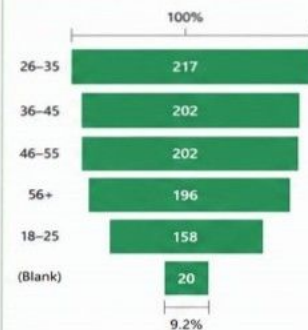
TOTAL CUSTOMERS BY CITY



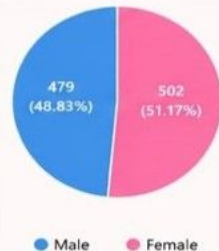
CUSTOMER DISTRIBUTION BY CITY & CATEGORY



TOTAL CUSTOMERS BY AGE GROUP



TOTAL CUSTOMERS BY GENDER



KEY INSIGHTS

- ▶ Patna has the highest number of customers (134) followed closely by Kolkata (132) and Mumbai (130).
- ▶ Electronics is the most contributing category in almost all cities.
- ▶ The 26-35 age group is the largest customer segment with 217 customers.
- ▶ Gender distribution is almost equal with a slight female majority (51.17%).
- ▶ Some records have missing city information, indicating incomplete data.



KEY FINDINGS

- ✓ Top customer markets: Patna, Kolkata, and Mumbai.
- ✓ Core customer age segment: 26-35 years.
- ✓ Gender split is balanced, with a slight female dominance.
- ✓ Electronics appears to be the most consistently strong category across cities.
- ✓ A small portion of records has missing city information, which may affect location-based analysis.



CONCLUSION

The business has a well-distributed customer base across major cities, age groups and genders. The 26-35 age group and the electronics category are the strongest segments. Overall gender distribution is balanced, which is a positive sign for business stability. Improving data quality for missing city records will help in more accurate analysis and decision making.



RECOMMENDATIONS

- ✓ Focus more marketing efforts on the 26-35 age group.
- ✓ Continue leveraging electronics as the top performing category.
- ✓ Run targeted campaigns in cities with lower customer count like Pune and Gaya.
- ✓ Improve data collection processes to reduce missing location records.



KEY MESSAGE:

A balanced customer distribution across cities, age groups and gender, with strong performance in the electronics category, indicates a healthy customer base and opportunities for further growth.



HYPOTHESIS TESTING SUMMARY



TO ANALYSE SALES DATA AND EVALUATE RELATIONSHIPS AND DIFFERENCES WITHIN THE DATASET.



1. T-TEST SALES BY GENDER

Independent Sample T-test



PURPOSE

To determine whether the average sales of male customers and female customers are significantly different.



HYPOTHESES

- H_0 (Null Hypothesis): There is no significant difference between male and female customer sales.
- H_1 (Alternative Hypothesis): There is a significant difference between male and female customer sales.



RESULT

The p-value obtained from the t-test was 0.47338.



INTERPRETATION

Since the p-value is greater than 0.05, the null hypothesis is not rejected.



CONCLUSION

There is no statistically significant difference between the sales generated by male and female customers. This indicates that both genders contribute to sales in a fairly similar manner.



2. ANOVA: SALES ACROSS CATEGORIES

One-way ANOVA



PURPOSE

To check whether average sales differ significantly across different product categories.



HYPOTHESES

- H_0 (Null Hypothesis): The average sales of all product categories are equal.
- H_1 (Alternative Hypothesis): At least one product category has a different average sales value.



RESULT

The ANOVA test showed a very small p-value (approximately 0).



INTERPRETATION

Since the p-value is less than 0.05, the null hypothesis is rejected.



CONCLUSION

There is a statistically significant difference in sales across product categories. This means that some categories perform better than others and product category has a strong impact on sales performance.



3. CHI SQUARE TEST: GENDER VS CATEGORY

Chi-square Test of Independence



PURPOSE

To determine whether customer gender and product category preference are associated.



HYPOTHESES

- H_0 (Null Hypothesis): Gender and product category are independent.
- H_1 (Alternative Hypothesis): Gender and product category are associated.



RESULT

The p-value obtained from the Chi-Square Test was 0.05838.



INTERPRETATION

Since the p-value is greater than 0.05, the null hypothesis is not rejected.



CONCLUSION

There is no statistically significant association between customer gender and product category. This suggests that category preference is not strongly dependent on gender in this dataset.



KEY MESSAGE

The analysis shows that sales performance varies significantly across product categories, while gender has no significant impact on sales or category preference. This highlights the importance of focusing on product categories to drive business growth.



HYPOTHESIS TESTING SUMMARY



To analyse sales data and evaluate relationships and differences within the dataset



OBJECTIVE

The purpose of hypothesis testing was to examine whether there are significant differences and relationships within the sales dataset by using three statistical tests: t-test, one-way ANOVA, and chi-square test.



1. T-TEST (MALE VS FEMALE SALES)

Objective: To compare the average sales of male and female customers.

H₀ (Null Hypothesis): There is no significant difference in mean sales between male and female customers.

H₁ (Alternative Hypothesis): There is significant difference in mean sales between male and female customers.

P-VALUE

0.4734

CONCLUSION

Since the p-value is greater than 0.05, the null hypothesis is not rejected. This means there is no statistically significant difference in average sales between male and female customers.

INTERPRETATION

Male and female customers do not differ significantly in their average sales contribution.



2. ONE-WAY ANOVA (SALES ACROSS PRODUCT CATEGORIES)

Objective: To determine whether average sales differ across the five product categories.

H₀ (Null Hypothesis): The mean sales of all product categories are equal.

H₁ (Alternative Hypothesis): At least one product category has a different mean sales value.

P-VALUE

0.5803

F-STATISTIC: 0.7171

F-CRITICAL: 2.3809

CONCLUSION

Since the p-value is greater than 0.05, and the F-statistic is less than the F-critical value, the null hypothesis is not rejected. Therefore, there is no statistically significant difference in average sales across product categories.

INTERPRETATION

Average sales across product categories are not significantly different.



3. CHI-SQUARE TEST (GENDER VS PRODUCT CATEGORY)

Objective: To test whether customer gender and product category are associated.

H₀ (Null Hypothesis): Customer gender and product category are independent.

H₁ (Alternative Hypothesis): Customer gender and product category are associated.

P-VALUE

0.0587

CONCLUSION

Since the p-value is greater than 0.05, the null hypothesis is not rejected. This indicates that there is no statistically significant association between customer gender and product category.

INTERPRETATION

Customer gender and product category are not strongly associated.



OVERALL CONCLUSION

The hypothesis testing results suggest that male and female customers do not differ significantly in their average sales, average sales across product categories are not significantly different, and customer gender and product category are statistically independent.

FINAL INSIGHT

Overall, the statistical tests indicate that sales performance is relatively balanced across gender and categories, and no strong statistical evidence was found for major differences or associations within the tested variables.

STATISTICAL TEST RESULTS

A. ANOVA TEST

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Column 1	354	50778582	143429.2	1.38E+10
Column 2	156	19621996	125783.3	1.11E+10
Column 3	159	21521561	135355.7	1.22E+10
Column 4	153	22231721	145300.1	1.32E+10
Column 5	178	25031689	140627.5	1.37E+10

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	2.74E+10	4	6.85E+09	0.717117	0.580286	2.380876
Within Groups	9.53E+12	995	9.58E+09			
Total	9.56E+12	999				

B. CHI SQUARE TEST

GENDER	ELECTRONICS	FASHION	FURNITURE	GROCERY	EDUCATION	ROW TOTAL
MALE	195	67	85	81	82	511
FEMALE	158	89	74	70	96	489
COLUMN TOTAL	354	156	159	151	178	1000

EXPECTED FREQUENCY TABLE

GENDER	ELECTRONICS	FASHION	FURNITURE	GROCERY	EDUCATION	ROW TOTAL
MALE	180.89	79.72	81.25	77.18	90.96	511
FEMALE	173.11	76.28	77.75	74.82	87.04	489

CHI-SQUARE STATISTIC (χ^2): 0.058680515

C. T TEST

$\sum x$	=SUMIF(I2:I512,"MALE",L2:L512)	72527094.8
$\sum x$	=SUMIF(I513:I1001,"Female",L513:L1001)	66872344.9
$\sum x$	=T.TEST(I2:P512,Q2:Q490,2,3)	0.47338

Legend:

SS: Sum of Squares | df: Degrees of Freedom | MS: Mean Square
F: F-statistic | P-value: Probability value | F crit: Critical F value